

Offshore DL Dissertation Results Snapshot – May 2026

Generated from CDF-gated readiness artifacts

2026-05-15

Scope

This snapshot summarizes current dissertation-facing benchmark evidence at source revision [62780b2dd5aa](#). Stage 1 3W, Stage 2 3W, Ganymede, and CDF results are reported with family-specific caveats. CDF trained reconstruction and foundation forecast metrics are not pooled.

3W Stage 1: standard 720-window HPO

Model	Macro-F1	Accuracy
Random Forest	0.968972	0.970228
DeepONet	0.962579	0.968734
MambaSL	0.962185	0.967313
LSTM	0.960046	0.965314
FKMAD	0.956388	0.964061
ConvTimeNet	0.954953	0.959894
PatchTST	0.953556	0.958738

3W Stage 2: separate follow-ups

Variant	Macro-F1	Accuracy
window360_rf	0.987797	0.991537
window1440_rf	0.977011	0.989084
wavelet_rf	0.964309	0.966301
physics_rf	0.964070	0.966060

Window variants use different window lengths/sample counts and are not Stage 1 replacements.

Ganymede post-fix forecasting

Model	MAE	RMSE	MASE
TiRex	0.3617	1.2476	0.2071
TimesFM	0.3965	1.2634	0.2295
Chronos-2	0.5357	1.4412	0.3205
LSTM	0.5457	1.3517	0.0228
TCN	0.5677	1.3136	0.0234
DeepONet	0.6795	1.3746	0.0301
PatchTST	1.0771	2.1164	0.0474

Foundation models lead by MAE/RMSE; trained LSTM/TCN lead by grouped MASE.

CDF post-fix anomaly detection

Trained reconstruction metrics

Model	error_mean	error_p50
lstm	0.005878	0.005289
patchtst	0.081621	0.071603
deeponet	0.230735	0.223694

Foundation forecast metrics

Model	forecast_error_mean	forecast_error_p50
chronos	0.243968	0.226866
tirex	0.268548	0.246441
timesfm	0.295999	0.283747

These CDF groups use different anomaly semantics and must not be pooled into one ranking.